

Department Overview Department Name: Department of Computer Engineering Faculty: Faculty of Information & Communication Technology (FICT) Accreditation: The program is fully accredited by the Pakistan Engineering Council (PEC). Core Focus: Blends hardware engineering and computer science concepts. It covers software programming, hardware design, industrial automation, embedded systems, robotics, multimedia, and artificial intelligence.

Vision and Mission Vision: To produce professional Computer Engineers who can use knowledge, technical expertise, and entrepreneurial abilities to compete with upcoming challenges in computing fields while adhering to highest standards of morality and ethics.

Mission: To prepare graduates to accept and fulfill responsibilities across a broad spectrum of activities, including analysis, synthesis and design of the modern computing systems, serve government and industry in various roles and contribute in academia and research.

Bachelor of Science (BS) Program

Admission Requirements Academic Background: F.Sc (Pre-Engineering) / ICS with Mathematics & Physics, or an equivalent qualification with at least 60% marks. Alternatively, a Diploma of Associate Engineering (DAE) in a relevant field with at least 60% marks. Entry Test: Candidates must clear an entrance test conducted by NTS.

Degree Requirements Total Credit Hours: 134 Total Courses: 43 Total Labs: 21 or 26 Minimum Academic Standing: CGPA $\geq$ 2.0

Curriculum Plan (Program Schema by Semester)

1st Semester • CS 118/118L | Computer Fundamentals & Lab | Prerequisite: None | Credit Hours: 2+1 • PHY 205/205L | Applied Physics & Lab | Prerequisite: None | Credit Hours: 2+1 • MATHP 118 | Calculus & Analytical Geometry | Prerequisite: None | Credit Hours: 3 • HUM 101 | Islamic Studies/Ethics | Prerequisite: None | Credit Hours: 2 • ENG 161 | Reading and Writing Skills (Eng.-1) | Prerequisite: None | Credit Hours: 3 • ENGG 202L | Workshop Practice Lab | Prerequisite: None | Credit Hours: 1 • ENVIRON102 | Occupational Health and Safety | Prerequisite: None | Credit Hours: 1

2nd Semester • CS 112/112L | Computer Programming & Lab | Prerequisite: None | Credit Hours: 3+1 • MATHP 111 | Linear Algebra | Prerequisite: None | Credit Hours: 2 • EE 221/221L | Electronic Circuits and Devices & Lab | Prerequisite: None | Credit Hours: 3+1 • ENG 361 | Communication Skills | Prerequisite: None | Credit Hours: 2 • EE 203/203L | Linear Circuit Analysis & Lab | Prerequisite: None | Credit Hours: 3+1

3rd Semester • HUM 102 | Pakistan Studies and Global Perspective | Prerequisite: None | Credit Hours: 2 • TE 305/305L | Computer Comm. and Networks & Lab | Prerequisite: None | Credit Hours: 3+1 • CS 212/212L | Object-Oriented Programming & Lab | Prerequisite: Computer Programming + Lab | Credit Hours: 3+1 • MATHA 213 | Complex Variables and Transforms | Prerequisite: None | Credit Hours: 3 • EE 110/110L | Digital Logic Design & Lab | Prerequisite: None | Credit Hours: 3+1 • MATHA 234 | Discrete Structures | Prerequisite: None | Credit Hours: 3

4th Semester • EE 261/261L | Signals & Systems & Lab | Prerequisite: Complex Variables and Transforms | Credit Hours: 3+1 • CE 211/211L | Computer Architecture and Organization & Lab | Prerequisite: Digital Logic Design + Lab | Credit Hours: 3+1 • MATHA 214 | Differential Equations | Prerequisite: Calculus & Analytical Geometry | Credit Hours: 3 • CS 214/214L | Data Structures & Algorithms & Lab | Prerequisite: Object-Oriented Programming + Lab | Credit Hours: 3+1 • TBA |

MS-Elective I | Prerequisite: None | Credit Hours: 3

5th Semester • CE 326/326L | Microprocessors and Interfacing & Lab | Prerequisite: Computer Architecture and Org. + Lab | Credit Hours: 3+1 • CS 341/341L | Operating Systems & Lab | Prerequisite: Object-Oriented Programming + Lab | Credit Hours: 3+1 • TBA | Social Science-Elective I | Prerequisite: None | Credit Hours: 2 • HUM 292 | Technical Writing | Prerequisite: None | Credit Hours: 3 • TE 412/412L | Digital Signal Processing & Lab | Prerequisite: Signals & Systems + Lab | Credit Hours: 3+1

6th Semester • TBA | CEDE - I & CEDE - II | Prerequisite: TBA | Credit Hours: Var (2+1 or 3) • MATHA 217 | Probability Methods in Engineering | Prerequisite: None | Credit Hours: 3 • CS 331/331L | Database Management Systems & Lab | Prerequisite: None | Credit Hours: 3+1 • EE 302/302L | Digital System Design & Lab | Prerequisite: Digital Logic Design + Lab | Credit Hours: 3+1

7th Semester • TBA | MDEE - I & CEDE-III | Prerequisite: TBA / None | Credit Hours: Var • TBA / TBA | Numerical Analysis & Lab | Prerequisite: TBA | Credit Hours: 2+1 • CE 451L | Senior Design Project-I | Prerequisite: None | Credit Hours: 3 • CE 303/303L | Software Engineering & Lab | Prerequisite: None | Credit Hours: 2+1

8th Semester • TBA | CEDE - IV | Prerequisite: TBA | Credit Hours: Var • TBA | Management Science Elective-II | Prerequisite: None | Credit Hours: 3 • TBA | MDEE - II | Prerequisite: None | Credit Hours: 3 • CE 452L | Senior Design Project-II | Prerequisite: Senior Design Project-I | Credit Hours: 3 • TBA | Social Science Elective-II | Prerequisite: None | Credit Hours: 3

Master of Science (MS) Program

Admission Requirements Academic Background: 16 years of education or equivalent (e.g., B.E or BS 4-year degree) in a relevant field from an HEC-recognized university. Applicants must have at least 60% marks under the annual grading system or a minimum CGPA of 2.5 out of 4.0. Entry Test: Must clear the admission test conducted by the University or its authorized testing agency with a minimum cumulative score of 50%.

Degree Requirements • Total Credit Hours: 30 • Total Courses: 10 • Minimum Academic Standing: CGPA $\geq$ 3.0